

Experiment No: 5

Title

Sqoop Tool Installation, Configuration and Selecting Database *Sqoop_DB* and Viewing Records from *Student Table*

Aim

To install and configure **Apache Sqoop** and to select the database **Sqoop_DB** and view the raw records from the **Student** table.

Software Requirements

- **Apache Hadoop**
- **Apache Sqoop**
- **MySQL**
- Java JDK
- Windows OS

Procedure

Step 1: Start Hadoop Services

Go to Hadoop sbin folder:

```
cd C:\hadoop\sbin
start-dfs.cmd
start-yarn.cmd
```

Verify services:

```
jps
```

Step 2: Verify Sqoop Installation

```
sqoop version
```

Step 3: List Databases from MySQL using Sqoop

```
sqoop list-databases --connect jdbc:mysql://localhost:3306/ --username root -  
-password ""
```

Step 4: Select Database Sqoop_DB and List Tables

```
sqoop list-tables --connect jdbc:mysql://localhost:3306/sqoop_db --username  
root --password ""
```

Step 5: View RAW Records from Student Table

```
sqoop eval --connect jdbc:mysql://localhost:3306/sqoop_db --username root --  
password "" --query "SELECT * FROM student"
```

Sample Output

i d	name	dep t
1	Arun	CS E
2	Bala	IT
3	Chitr a	AI

Result

Thus the **Sqoop Tool** was successfully installed and configured, and the database **Sqoop_DB** was selected and raw records from the **Student** table were viewed successfully using Sqoop.

Experiment No: 6

Title

Importing MySQL Table Data into HDFS using Sqoop Tool

Aim

To import the data from MySQL **Student** table in **Sqoop_DB** database into HDFS using **Apache Sqoop**.

Software Requirements

- **Apache Hadoop**
- **Apache Sqoop**
- **MySQL**
- Java JDK
- Windows OS

Procedure

Step 1: Start Hadoop Services

```
cd C:\hadoop\sbin  
start-dfs.cmd  
start-yarn.cmd
```

Verify:

```
jps
```

Step 2: Create Directory in HDFS

```
hdfs dfs -mkdir /sqoop_import
```

Step 3: Import Student Table from MySQL to HDFS

```
sqoop import --connect jdbc:mysql://localhost:3306/sqoop_db --username root -  
-password "" --table student --target-dir /sqoop_import --m 1
```

Command Explanation

- --table student → MySQL table name
- --target-dir → HDFS destination
- --m 1 → number of mappers

Step 4: View Imported Data in HDFS

```
hdfs dfs -ls /sqoop_import  
hdfs dfs -cat /sqoop_import/part-m-00000
```

Sample Output

```
1,Arun,CSE  
2,Bala,IT  
3,Chitra,AI
```

Result

Thus the data from MySQL **Student** table was successfully imported into HDFS using Sqoop.

Experiment No: 7

Title

Exporting Data from HDFS to MySQL Table using Sqoop Tool

Aim

To export the data from HDFS directory into MySQL **Student_Export** table in **Sqoop_DB** database using **Apache Sqoop**.

Software Requirements

- **Apache Hadoop**
- **Apache Sqoop**
- **MySQL**
- Java JDK
- Windows OS

Procedure

Step 1: Start Hadoop Services

```
cd C:\hadoop\sbin  
start-dfs.cmd  
start-yarn.cmd
```

Verify services:

```
jps
```

Step 2: Create Table in MySQL

Login to MySQL:

```
mysql -u root -p
```

Select database:

```
USE sqoop_db;
```

Create table:

```
CREATE TABLE student_export(  
id INT,  
name VARCHAR(20),  
dept VARCHAR(20)  
);
```

Step 3: Export Data from HDFS to MySQL

```
sqoop export --connect jdbc:mysql://localhost:3306/sqoop_db --username root -  
-password "" --table student_export --export-dir /sqoop_import --m 1
```

Step 4: Verify Exported Data in MySQL

```
SELECT * FROM student_export;
```

Sample Output

id name dept

1 Arun CSE

2 Bala IT

3 Chitra AI

Result

Thus the data was successfully exported from HDFS to MySQL table using Sqoop.